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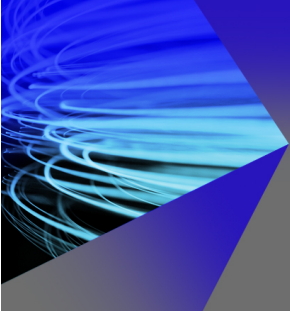
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ABSTRACT

The effect of a pair of synthetic jets (SJs) on sound generation from a circular cylinder undergoing vortex-induced vibration (VIV) is numerically investigated under two-dimensional conditions. The cylinder is immersed in a uniform flow and vibrates in the transverse direction at a Reynolds number of 150 and a Mach number of 0.05. Various frequencies (f_{sj}^*) and phase differences ($\Delta\phi$) of SJs are employed to examine their impact on far-field sound radiation. The modulation of VIV and flow structure by SJs also affect the sound generation and propagation. Anti-phase SJs can sufficiently suppress the intensity of the lift dipole and the drag dipole by 85% and 75%, respectively, while the drag dipole dominates the sound field. Through the dynamic mode decomposition analysis for the spatial distribution of components associated with various frequencies, the sound generation mechanism is revealed. The adverse effect of SJs has been attributed to the amplification of the harmonic component and self-noise generated from SJs due to a non-zero mass flow rate or injection of momentum flux. And combining with the acoustic analogy equation, these mechanisms of sound generation are first determined in two-dimensional conditions. Consequently, SJs can efficiently reduce the sound radiation by suppressing VIV. However, their disadvantages, including self-noise generated by SJs and the increased high-frequency components, have also been emphasized.

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I. INTRODUCTION

Vortex-induced vibration (VIV) is a typical flow-structure interaction phenomenon, which is widely seen in engineering.^{1–3} The cylinder with an elastic support may vibrate under the excitation of vortex shedding. Especially when the vortex shedding frequency is close to the natural frequency, the vibration amplitude will significantly augment, and this phenomenon is called “lock-in.” Due to the occurrence of lock-in, our previous work⁴ reported that the more intense noise was generated compared to that from the stationary circular cylinder. While numerous control strategies exist for the flow past a stationary cylinder, noise reduction in fluid-structure interaction systems remains underdeveloped due to the complicated noise generation mechanisms. Therefore, it is vital to find an efficient control approach for reducing noise generated from the vortex-induced vibrating cylinder.

It is well known that the vibration of the structure in a stationary medium can generate noise.⁵ In addition, from an aeroacoustic perspective, the unsteady flow also induces various sound sources. As for the flow past a stationary circular cylinder, Inoue and Hatakeyama⁶

utilized two-dimensional direct simulation in the laminar condition to investigate the mechanism of the sound generation. The result indicates that pressure pulses are induced by vortex shedding from the upper and lower sides of the cylinder and propagate to the far field. For the cylinder undergoing VIV in the transverse direction, we found that the vortex shedding and flow separation were affected by the movement of the cylinder, which led to the increase in the hydrodynamic fluctuation and ultimately an increase in the overall radiated noise. The far-field noise was still dominant by the lift dipole, but the drag dipole was significantly enhanced in comparison to the stationary cylinder.⁴ The characteristic of noise radiation can be an analogy by forcing the cylinder to oscillate, which is a reasonable approach for investigating VIV problems.⁷ Effects of the forced oscillation control on sound generation from the cylinder have been discussed by direct⁸ and hybrid⁹ acoustic approaches, respectively. Similar characteristics are demonstrated: the lift dipole dominates the far-field noise, but the drag dipole has also been significantly increased. However, the noise radiation was significantly modified by the oscillation frequency and amplitude under the influence of unsteady hydrodynamic and

cylinder's movement. In the present study, it indicates that the abundant noise generation mechanisms in fluid–structure interaction problems are the key issues of sound control.

To reduce noise radiation, numerous passive and active flow control methods, such as porous coating,¹⁰ splitter plates,¹¹ serrated trailing edge,¹² noise plasma actuators,^{13,14} rotary oscillation,^{15–19} pure blow,^{20–22} and surface perturbation,²³ have been implemented. Although the active flow control requires external energy input, which might introduce extra sound sources, it is still a major area of interest for noise reduction. We summarized that the application of active flow control to noise reduction includes two approaches. One is to modulate the noise by the interference between the self-noise of actuators and the original noise in the process of sound propagation. Inoue *et al.*²⁴ demonstrated a control strategy employing periodic blowing/suction jets positioned on the upper and lower surfaces of the cylinder, and these jets were configured to introduce an additional monopole source. To avoid affecting the vortex shedding or the amplitude of lift force, the jet intensity was maintained at a low level. The numerical results showed that both the sound generation and propagation can be modified without drastic modification of the flow field. Guo²⁵ also proposed a low-noise design for array actuators using modulated time delay inspired by the concept of noise cancelation between jets, which was applicable to low Mach number flows and smooth geometries such as airfoil surfaces. As validated by the numerical simulation, this design achieved significant reductions in near-field pressure fluctuations and far-field noise without compromising flow control effectiveness. The above-mentioned studies are based on an idealized assumption that the actuation produces only additional sound sources and its effect on the flow is ignored. However, the effects of flow and noise have to be considered simultaneously to improve the robust when the application scenario becomes more complex.

The other is based on the control of flow field, and the noise reduction is achieved by modulating the generation of sound sources. Ganta *et al.*¹⁸ numerically investigated the effect of rotary oscillation on the circular cylinder's sound field in the laminar condition. However, the noise reduction was a failure in the inflow direction. It was worth noting that external perturbation also changed the wavelengths of sound waves. Du and Sun¹⁷ employed a hybrid approach to reduce flow-induced noise via closed-loop rotary oscillation of a circular cylinder at $Re = 1000$ and $Ma = 0.2$. Feedback rotary oscillation significantly suppressed vortex shedding, mitigating lift fluctuations by over 50% and achieving up to 10 dB noise reduction. As for the flow-induced vibration, Cheng and Zhou²³ implemented a surface perturbation technique to generate localized vibrations on structures. The experimental results demonstrated that closed-loop control outperformed open-loop control and successfully suppressed the noise by adjusting the phase between fluctuating pressure and structure vibration. Liu *et al.*²⁶ utilized the rotary oscillation for the circular cylinder with a nonlinear elastic support at $Re = 150$ and $Ma = 0.2$. An amplification of the harmonic component was found in far-field sound radiation. Dynamic mode decomposition (DMD) analysis revealed sound fields dominated by lock-on frequencies, while complex multi-frequency interactions exhibited in the non-lock-on regime. These active flow control investigations have shown the efficient suppression of hydrodynamic fluctuation, but they have also demonstrated that this suppression is not directly equivalent to noise reduction. Specifically, the self-noise generated by actuators (e.g., momentum flux

injection) and the potential amplification of harmonic components may weaken noise reduction, leaving a critical challenge for noise control.

Utilizing high-momentum fluid to modulate transverse flow movement in the vortex-formation region is thought to be a key factor for reducing noise. Based on this concept, sound generation from the stationary cylinder has been reduced by blowing control in the anechoic wind tunnel.^{21,22} As a more flexible approach, synthetic jets (SJs) control has been used in the drag reduction and the suppression of VIV. Wang *et al.*²⁷ investigated the effect of control parameters on wake patterns. The frequency and phase difference between upper and lower SJs resulted in different structures' responses. Furthermore, effects of an important state called "lock-on" on control performance were comprehensively discussed.²⁸ In the lock-on regime, the vortex shedding frequency was affected by SJs and shifted away from the natural frequency. Lock-on regime in the present study refers specifically to the vortex synchronization that occurs under periodic external excitation,²⁹ which is different from the lock-in regime in the VIV.¹ Ren *et al.*³⁰ utilized jets with the control laws chosen by genetic programming to significantly suppress VIV. More complex goals, such as hydrodynamic stealth³¹ and enhancing heat transfer,³² have become possible with the advancement of artificial intelligence.^{33,34} While these active flow control approaches have demonstrated efficacy in drag reduction or VIV suppression, their acoustic impacts remain poorly understood. A realistic application is when using SJs to suppress the VIV of offshore riser, the affect of SJs-induced noise on underwater systems and marine ecosystems has to be considered.

This study aims to give a comprehensive assessment of synthetic jets (SJs) for controlling sound generated from a circular cylinder undergoing VIV. We mainly focus on the modification of SJs on the acoustic characteristics in the far-field, including the intensity, the frequency, and the acoustic radiation directivity. Through high-fidelity simulations combining the generalized formulation of the interpolation-supplemented cascaded lattice Boltzmann method (GICLBM) and mature analytical approaches (e.g., the multi-pole decomposition and DMD analysis), we reveal how SJs' frequency and phase difference govern the sound radiation. Based on the acoustic analogy theory, the sound generation mechanism is further analyzed. This work will provide new insights for using SJs to simultaneously suppress VIV and modulate far-field sound radiation.

II. PROBLEM DESCRIPTION AND METHODOLOGY

A. Problem description

A circular cylinder is immersed in a uniform flow. The Mach number is defined as $Ma = U_0/c_0$, where U_0 is the velocity of a uniform flow and c_0 is the speed of sound. Controlling the sound radiation of the circular cylinder undergoing VIV is in the laminar condition, i.e., $Re = 150$. The Reynolds number is calculated by $Re = U_0 D_0/\nu_0$, where D_0 is the diameter of the cylinder and ν_0 is the kinematic viscosity. Flow past the circular cylinder produces vortex shedding, which results in the force exerting on the cylinder in periodical variation. Since the cylinder is connected to a spring as shown in Fig. 1, the downstream vortex shedding can induce the cylinder to transversely vibrate. The vibration is constrained in the transverse direction, and the displacement of the cylinder $y(t)$ is given by

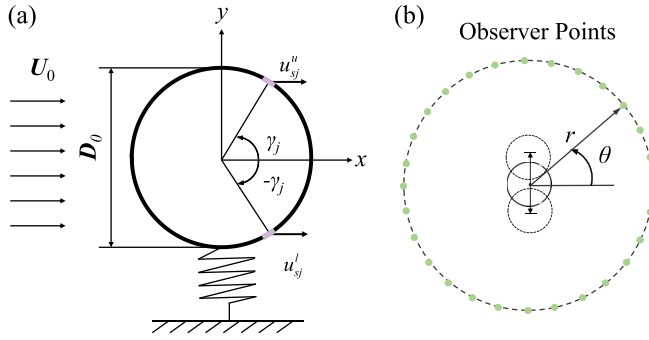


FIG. 1. A pair of SJs blowing/suctioning from a transversely vibrating cylinder is shown schematically. U_0 is the velocity of uniform flow and D_0 is the diameter of the cylinder. Purple lines represent SJs, which are located on the upper and lower leeward sides of the cylinder. $u_{sj,u}$ and $u_{sj,l}$ are SJs' velocities. γ is the azimuthal angle. K is the spring stiffness coefficient. 256 observer points are located in the far field to measure the fluctuating pressure.

$$m \frac{\partial^2 y}{\partial t^2} + Ky = F_L, \quad (1)$$

where m is the mass of cylinder, K is the spring stiffness coefficient, and F_L is the lift force on cylinder's surface. The natural frequency of this mass-spring system can be calculated as

$$f_n = \frac{1}{2\pi} \sqrt{\frac{K}{m}}. \quad (2)$$

The mass ratio m^* is defined as

$$m^* = \frac{4m}{\rho_0 D_0^2 \pi}. \quad (3)$$

Meanwhile, the reduced velocity U_R is expressed as

$$U_R = \frac{U_0}{f_n D_0}. \quad (4)$$

In the present study, the mass ratio and reduced velocity are chosen as $m^* = 2.55$ and $U_R = 5.9$.²⁸ Considering the effect of the added mass, F_L can be divided into two components: potential added-mass force (F_A) and vortex force (F_V), which can be calculated as²⁸

$$F_L = F_A + F_V, \quad (5)$$

and the potential added-mass force is

$$F_A = -C_A m \ddot{y}, \quad (6)$$

where C_A is the added-mass coefficient and assumed to be 1.0.³⁵ As a result, F_A is always in phase with $y(t)$. When the cylinder is undergoing VIV, the lift force includes three components: spring tension, potential added-mass force from external fluid due to transverse movement, and the vortex force from vortex shedding. In particular, the force in the transverse direction is represented by the sum of potential added-mass force and vortex force. The forces are all normalized by $\frac{1}{2} \rho_0 U_0^2 D_0$.

Considering the effect of the added mass, Eq. (1) is developed to

$$(m + C_A \rho_0 D_0^2) \frac{\partial^2 y}{\partial t^2} + Ky = F_V. \quad (7)$$

The natural frequency in Eq. (2) can also be developed to

$$f_n = \frac{1}{\pi} \sqrt{\frac{K}{4m + C_A \rho_0 D_0^2 \pi}}. \quad (8)$$

The frequency and amplitude of vibrations are normalized by

$$A^* = y/D_0, \quad f^* = f/f_n. \quad (9)$$

As shown in Fig. 1, a pair of SJs are arranged symmetrically about the centerline, $y = 0$, on the leeward surface of the cylinder. The letter γ_j is the azimuthal angle of SJs and located at $\gamma_j = 70^\circ$, which is near the separation point of the stationary cylinder, to facilitate a more effective interaction with the shear layer. To avoid directly applying external force in the vibrating direction, SJs act in the same direction as the incoming velocity. The instantaneous SJs' velocities are given by

$$\begin{aligned} u_{sj,u}^u &= u_{sj} \cos(2\pi f_{sj} t + \phi_u), \\ u_{sj,l}^l &= u_{sj} \cos(2\pi f_{sj} t + \phi_l), \end{aligned} \quad (10)$$

where u_{sj} is the amplitude of SJs' velocity and f_{sj} is the frequency of SJs. ϕ_u and ϕ_l are the phases of upper and lower jets, respectively. The phase difference between upper and lower jets is defined as

$$\Delta\phi = \phi_u - \phi_l. \quad (11)$$

The cases of f_{sj}^* and $\Delta\phi$ are summarized in Table I.

In the present study, SJs are activated when $y(t)$ varies from negative to positive. The state of vibration is confirmed to be independent of initial conditions, i.e., SJs' starting timing.²⁸ The phase of upper jet ϕ_l is set to 0. While strong SJs are usually more effective for flow control, we used to find that more extra sound is generated from SJs,³⁶ which is also exhibited in the present study. To optimize the balance between achieving significant flow control performance and mitigating SJs' self-noise, u_{sj} is set to $4U_0$. The strength of SJs is calculated by a momentum coefficient $C_\mu = \frac{2u_{sj}^2 d_{sj}}{U_0^2 D_0}$, where d_{sj} is the length of a single jet. With setting $d_{sj} = \frac{\pi D_0}{64}$, C_μ is 1.57 in the present study.

B. Numerical methodology

The direct acoustic simulation for VIV involved two critical issues about flow-induced vibration and flow-induced noise. To facilitate this investigation, we developed the generalized formulation of the interpolation-supplemented cascaded lattice Boltzmann method (GICLBM).⁴ The GICLBM framework extends the classical lattice Boltzmann method (LBM) on non-uniform grids to enable sound prediction with a moving rigid-body. The following is a brief introduction to GICLBM, detailed implementation, and validation can be obtained from our previous work.^{4,37} The lattice Boltzmann method (LBM) is a powerful tool for solving the weakly compressible Navier-Stokes equations and has been extended to the multi-phase flow,³⁸ the non-Newtonian fluids,³⁹ and the heat transfer.⁴⁰ It is also considered a promising approach⁴¹ for computational aeroacoustics, thanks to the low-dissipation and low-dispersion features.⁴² The classical LBM, including collision and streaming processes, was implemented on a non-uniform grid through the second-order interpolation and the coordinate transformation.⁴³ In order to improve the stability of calculation, the cascaded model^{44–46} was also utilized for the collision

TABLE I. Summary of control parameters.

Parameters	Values
f_{sj}^*	0.5, 0.6, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.8, 1.9, 2.0, 2.1, 1.8, 1.9, 2.0, 2.1, 2.2, 2.5, 2.7 3.0, 3.1, 3.2, 4.0, 4.1, 5.0.
$\Delta\phi$	0, 0.25 π , 0.5 π , 0.75 π , π

process. The simulation was carried out on the moving frame of reference method to handle VIV.⁴⁷ In particular, to avoid the pollution due to sound waves reflected from the boundary of the computational domain, the perfectly matched layer (PML)⁴⁸ was adopted in GICLBM.

For obtaining the flow and sound fields simultaneously, the physical domain is divided into three regions and discriminated by a structured O-type grid in Fig. 2(a). The first region, $r < r_1$, is adjacent to the cylinder, and the finest mesh resolution is implemented to resolve the flow structure. The second region, $r_1 < r < r_2$, is where sound waves linearly propagate and the magnitude of sound generally decays with distance. The grid is uniformly stretched along the radial direction, and the mesh resolution should satisfy low dissipation and low dispersion in the sound propagation. It should be noted that more grid points are arranged compared with the classical LBM method to ensure that no obvious dispersion or dissipation occurs due to the use of second-order interpolation in GICLBM. In the third region, the progressively coarser grid is utilized as the PML layer.

The physical domain is close to $420D_0$ and includes a $103.5D_0$ PML layer. In GICLBM, the temporal resolution is limited by the CFL condition, and the present CFL number is set to 0.9. This numerical setup apparently requires more computational resources compared to the fluid simulation focusing on the near-field flow. For improving the computational efficiency, GICLBM is programmed in CUDA C language and implemented on a graphic processing unit (GPU). Moreover, an implicit streaming technology is also developed to reduce the memory usage.³⁷ Both the cylinder and SJs are realized by the specified velocity. The pressure in the far-field boundary is set by the reference pressure. These Dirichlet boundary conditions are handled by the non-equilibrium extrapolation scheme.⁴⁹

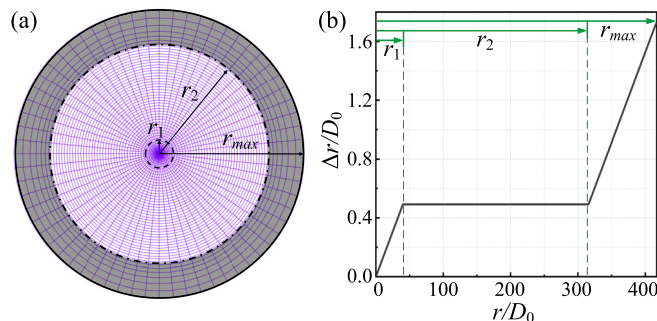


FIG. 2. (a) The structured O-type grid and (b) the node distribution along the radial direction. The PML layer is denoted by the gray region in (a), $r_2 < r < r_{max}$.

Since the cylinder moves in the transverse direction, the downstream wake patterns are arranged on different sides of the centerline, which are called “C2S” modes in Fig. 3(a). The vorticity is calculated by $\omega = 0.5 \times (\partial_x u_y - \partial_y u_x)$. The cylinder vibrates at the natural frequency leading to periodic variations of C_L , C_D , and A^* . The Strouhal number (St) is defined as $St = f_v D_0 / U_0$, where f_v is the vortex shedding frequency. The root mean square (RMS) values of the lift force ($C_{L,rms}$), the mean drag ($\bar{C}_D$), and St are all calculated for more than 30 cycles. After a comprehensive validation in Table II, the grid of $N_\theta \times N_r = 512 \times 1024$ is selected. The node distribution along the radial direction is depicted in Fig. 2(b). In the first region, $r_1 = 39.9D_0$ is implemented with a first wall-normal grid spacing of $0.006D_0$. In the second region for simulating the sound propagation, the grid is stretched at constant values $\Delta r = 0.49D_0$, as shown in Fig. 2(b). This spatial resolution ensures at least 215 grid points per wavelength (PPW) for the dominant frequency component ($f = f_n$) and effectively guarantees the accuracy in the concerned frequency range. As shown in Fig. 4(a), the decay of the maximum sound pressure p'_{max} is proportional to $1/\sqrt{r}$, which satisfies the theoretical prediction. Both p'_{max} and the instantaneous p' measured at $(0, 100D_0)$ are also consistent with the results of Inoue and Hatakeyama.⁶ It should be noted that all numerical simulation results except for instantaneous contours are exhibited in the absolute coordinate by the bilinear interpolation.

III. RESULTS AND DISCUSSION

A. Effect of SJs on flow fields undergoing VIV

When SJs are activated, the downstream flow field bifurcates into two distinct regimes. As illustrated in Fig. 5(b), the lock-on regime occurs when the vortex shedding frequency synchronizes with external perturbations induced by SJs, particularly when the SJs' frequency approaches the natural frequency. In contrast, the non-lock-on regime maintains a quasi-periodic flow structure in Fig. 5(c). The foundational

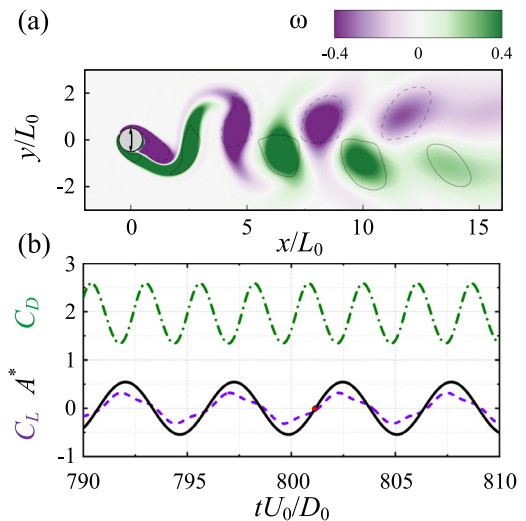


FIG. 3. (a) Instantaneous vorticity contour where positive and negative vorticities are enclosed by $\lambda_{ci} = 0.1$,⁵⁰ respectively. (b) Time histories of C_L , A^* , and C_D , which are represented by a purple dashed line, a black solid line, and a green dashed dotted line, respectively. The red circle marks the time instance corresponding to the vorticity contour.

TABLE II. Mesh convergence study and validation for the scenarios involving the fixed and vortex-induced vibrating cylinders.

Re	Case	No. of grid cells ($N_\theta \times N_r$)	Fixed cylinder			VIV cylinder	
			$\bar{C}_D$	$C_{L,rms}$	St	A_{max}^*	St
150	Present	256×512	1.201	0.299	0.171	0.555	0.191
		512×1024	1.280	0.345	0.182	0.544	0.192
		1024×2048	1.303	0.358	0.183	0.552	0.190
	Komatsu <i>et al.</i> ⁵¹	...	1.332	0.3643	0.1832	...	...
	Ma <i>et al.</i> ⁸	...	1.334	0.367	0.182	...	...
	Zhou <i>et al.</i> ⁵²	...	1.284	...	0.181	...	...
100	Present	512×1024	...	...	...	0.556	0.183
	Wang <i>et al.</i> ²⁷	...	...	...	0.169	0.545	0.186
	Ren <i>et al.</i> ⁵³		1.383	...	0.166	0.539	0.183

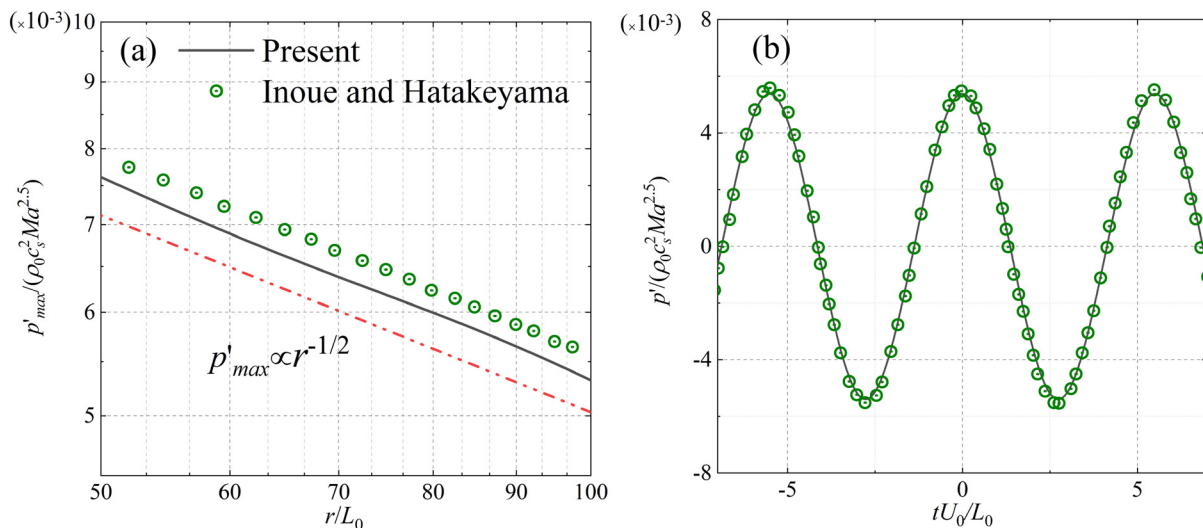
work by Wang *et al.*²⁷ systematically investigated the influence of SJs on the VIV, while Wang *et al.*²⁸ further elucidated the critical role of lock-on phenomena in modulating the VIV. One can refer to these previous works if interested. Building upon these insights, this study extends the investigation to the acoustic domain.

The variation of A^* with f_{sj}^* and $\Delta\phi$ is summarized in Fig. 6, where lock-on and non-lock-on regimes are represented by the solid and empty symbols, respectively. As defined by Wang *et al.*,²⁸ these lock-on cases are divided into four regions: primary, superharmonic, secondary, and tertiary, which correspond to vibrating frequencies of 1, 2, $1/2$, and $1/3$ of f_{sj}^* , respectively. It should be noted that a lock-on region would be identified only if SJs in it could shift the vibration frequency away from the natural frequency. Hence, even if $f_{sj}^* = 4$ and $f_{sj}^* = 5$, indeed, enter the periodic state, they are not classified into a lock-on region.

The vibration of the cylinder is an important sound source, and its suppression is a rational approach to reduce the far-field noise. As

shown in Fig. 6, A^* is lower than the uncontrolled case for most cases, indicating that the vibration has been successfully suppressed by SJs. In general, forcing the vibration frequency away from the natural frequency is one of the mechanisms to suppress VIV.²⁸ As for a single lock-on region with the same $\Delta\phi$, the farther away the vibration frequency is from the natural frequency, the smaller the amplitude is. For example, at $\Delta\phi = 0$, SJs with $f_{sj}^* = 1.3$ suppressed A_{rms}^* to 0.054, which is more effective compared to $A_{rms}^* = 0.180$ at $f_{sj}^* = 1.1$. Furthermore, even if SJs perform at the natural frequency (i.e., $f_{sj}^* = 1$), VIV is still alleviated. Another mechanism involved is to suppress the vortex strength.²⁷ Hence, there is no evidence proving that the occurrence of the lock-on regime necessarily leads to the suppression of VIV.

In addition to f_{sj}^* , the phase difference $\Delta\phi$ also affects the range of the lock-on region. As shown in Fig. 6, the primary and tertiary lock-on regimes occur at $\Delta\phi = \pi$, $\Delta\phi = 0.75\pi$, and $\Delta\phi = 0.5\pi$. In contrast, the secondary lock-on regime is at $\Delta\phi = 0.25\pi$ and $\Delta\phi = 0$. In the primary and tertiary lock-on regions, A_{rms}^* increases as $\Delta\phi$

**FIG. 4.** For flow past a stationary circular cylinder, (a) the decay of the maximum fluctuating pressure p'_{max} along $\theta = 90^\circ$ and the instantaneous fluctuating pressure measured at $(0, 100 D_0)$ at $Re = 150$ and $Ma = 0.05$. The green circles denote the results from Inoue and Hatakeyama.⁵

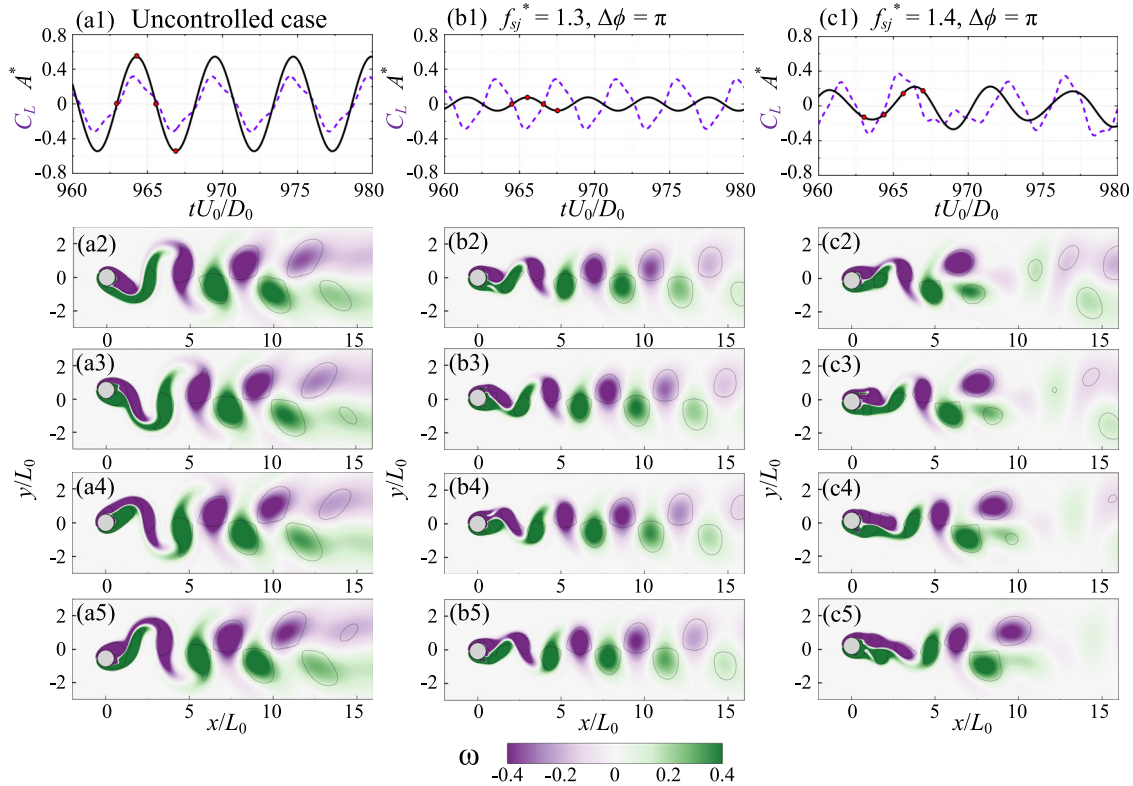


FIG. 5. Time histories of C_L and A^* , and instantaneous vorticity contours for (a) uncontrolled, (b) $f_{sj}^* = 1.3$, $\Delta\phi = \pi$, and (c) $f_{sj}^* = 1.4$, $\Delta\phi = \pi$ cases. C_L and A^* are represented by a purple dashed line and a black solid line, respectively. The red circle marks the time instance corresponding to the vorticity contour.

approaches π at the same f_{sj}^* . However, A_{rms}^* of $\Delta\phi = 0$ is slightly larger than that of $\Delta\phi = 0.25\pi$ in the secondary lock-on region. The superharmonic regime is more complicated, and the occurrence is at $\Delta\phi = 0.75\pi$, $\Delta\phi = 0.5\pi$, and $\Delta\phi = 0.25\pi$. The maximum A_{rms}^*

occurs at $f_{sj}^* = 0.5$, $\Delta\phi = 0.5\pi$, while it is the minimum with $f_{sj}^* = 0.6$ and $\Delta\phi = 0.25\pi$.

For the regime undergoing VIV, as shown in Fig. 3(a), the wake pattern exhibits spatiotemporal symmetry. The flow fields under SJs'

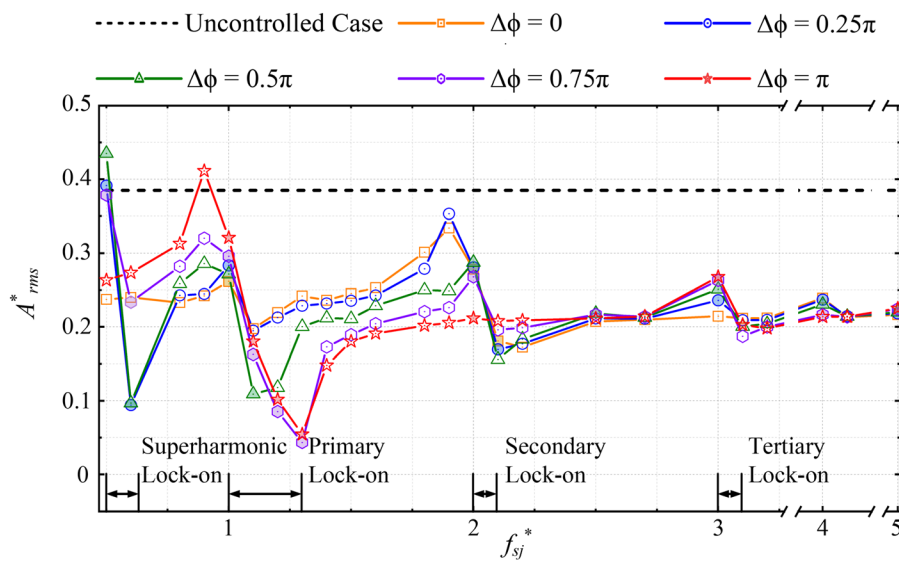


FIG. 6. Variation of A_{rms}^* with various f_{sj}^* and $\Delta\phi$. The case in the lock-on regime is denoted by the solid symbol, and the non-lock-on regimes by empty symbols. A_{rms}^* are calculated within more than 30 vibration cycles.

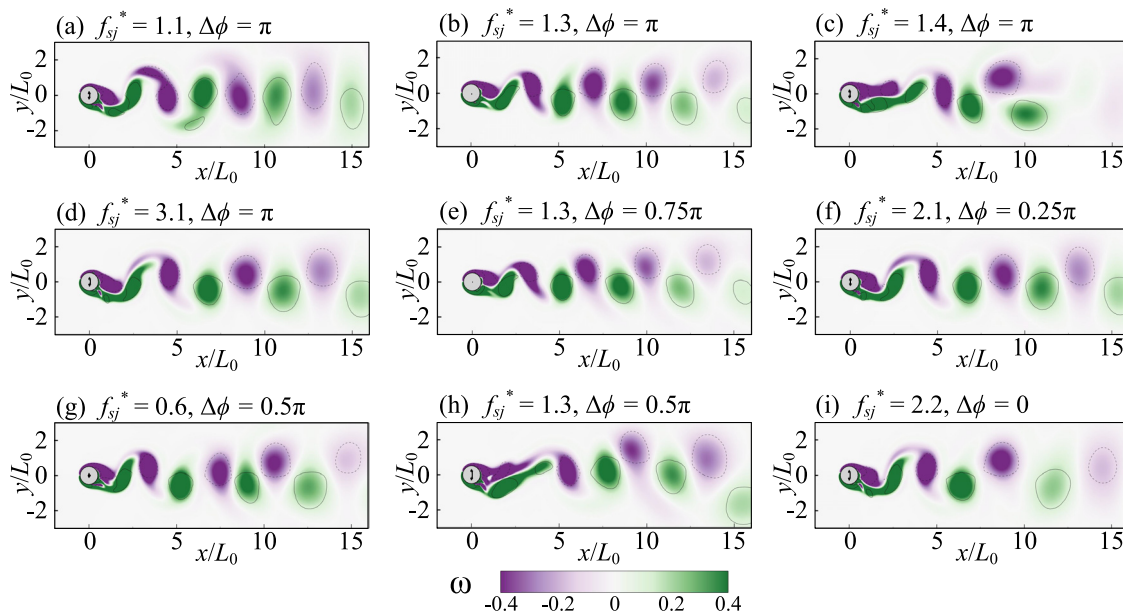


FIG. 7. Instantaneous vorticity contours at various f_{sj}^* and $\Delta\phi$, where the solid and dashed lines represent the region enclosed by $\lambda_{ci} = 0.1$.

control are depicted in Fig. 7, where Fig. 7(c), Fig. 7(h), and Fig. 7(i) are non-lock-on cases, and the rest is in the lock-on regime. Vortices in the lock-on regime alternatively shed from the cylinder, while those in the non-lock-on state tend to be quasi-periodic.

The combination and break of vortices in the adjacent region due to the sustained momentum injection from SJs can lead the downstream wake patterns to change. Because SJs do not change the vibration frequency for non-lock-on cases, the mismatch between SJs actuation period and vortex shedding period leads the downstream vortices to be stretched or pressed to different sizes. For lock-on cases, all of wake patterns are not altered by SJs and belong to the C2S pattern, but their quantities and strengths vary significantly. For example, there are more pairs of positive and negative vortices for $f_{sj}^* = 1.3$ than $f_{sj}^* = 1.1$ in Figs. 7(a) and 7(b), because the vortex shedding frequency is shifted in lock-on regime. However, although f_{sj}^* successively decreases in Figs. 7(d) $f_{sj}^* = 3.1$, 7(f) $f_{sj}^* = 2.1$, and 7(g) $f_{sj}^* = 0.6$, the quantity of vortices gradually increases. The reason is they belong to different lock-on regions, and the vortex shedding frequencies determined by the vibration frequency are $1/3$, $1/2$, and 2 of f_{sj} , respectively. The phase difference can also lead to the loss of spatiotemporal symmetry for the wake distribution. In Figs. 3(b) and 3(e), the anti-phase SJs maintain spatiotemporal symmetry, while the wake pattern deflects in the positive direction of the y -axis at $\Delta\phi = 0.75\pi$. Because the vortices generated by the upper and lower sides are typically anti-phase, it is reasonable to believe that anti-phase SJs do not break the spatiotemporal symmetry. Considering that the suction of SJs can weaken the shear layer but the blowing can enhance it, the vortex shedding may be delayed or promoted. Hence, SJs with $\Delta\phi = 0.75\pi$ lead the downstream wake pattern to shift from the centerline. This finding is consistent with the result of Wang *et al.*,²⁹ which contains a more detailed discussion.

In general, the dipole is generated from the hydrodynamic fluctuation for the stationary cylinder. To evaluate this fluctuation, the RMS values of hydrodynamics are calculated, and the variations of $C'_{L,rms}$ and $C'_{D,rms}$ are depicted in Fig. 8 with various f_{sj}^* and $\Delta\phi$. Researchers are usually concerned with the mean drag, but the fluctuation component is more critical from the acoustic perspective. Although SJs directly inject extra momentum fluctuation in the drag direction, $C'_{D,rms}$ is reduced compared to the unforced cylinder for almost all cases shown in Fig. 8, which is consistent with the trend of A^* . The suppression of drag fluctuation indicates that SJs can effectively reduce the sound generation in the drag direction. For non-lock-on cases, $C'_{D,rms}$ exhibits a decreasing trend as $\Delta\phi$ approaches 0 because the instantaneous momentum generated by anti-phase SJs undergoes cancellation. The drag fluctuation decreases as the vibration frequency deviates from the natural frequency in the lock-on regime for the same $\Delta\phi$. For non-lock-on cases with $f_{sj}^* < 2$ (excluding $\Delta\phi = \pi$), the lift fluctuation intensifies. When $f_{sj}^* > 2$, the lift fluctuation for non-lock-on cases decreases as $\Delta\phi$ approaches 0. Moreover, no clear relationship can be found between the mitigation of lift fluctuations and the onset of lock-on. The suppression of $C'_{L,rms}$ is not as obvious as that of A'_{rms} , e.g., $f_{sj}^* = 1.2$ and $\Delta\phi = \pi$. Different from the variation in Fig. 6, the extreme minimum value does not appear at the border of the lock-on region, e.g., the primary lock-on region with $\Delta\phi = \pi$. While the amplitude of the vibration is reduced by SJs, the vibration frequency is increased in the lock-on regime, which possibly leads to an augmentation of the potential added mass force in Eq. (6). These results highlight the effect of vibration and vortex shedding on sound generation should be considered simultaneously.

To demonstrate the mechanism for the suppression of VIV, the analytical solution for the undamped single-degree-of-freedom vibration system governed by Eq. (7) is derived with a simplification of the vortex force. If supposing the vortex force in the lock-on regime is

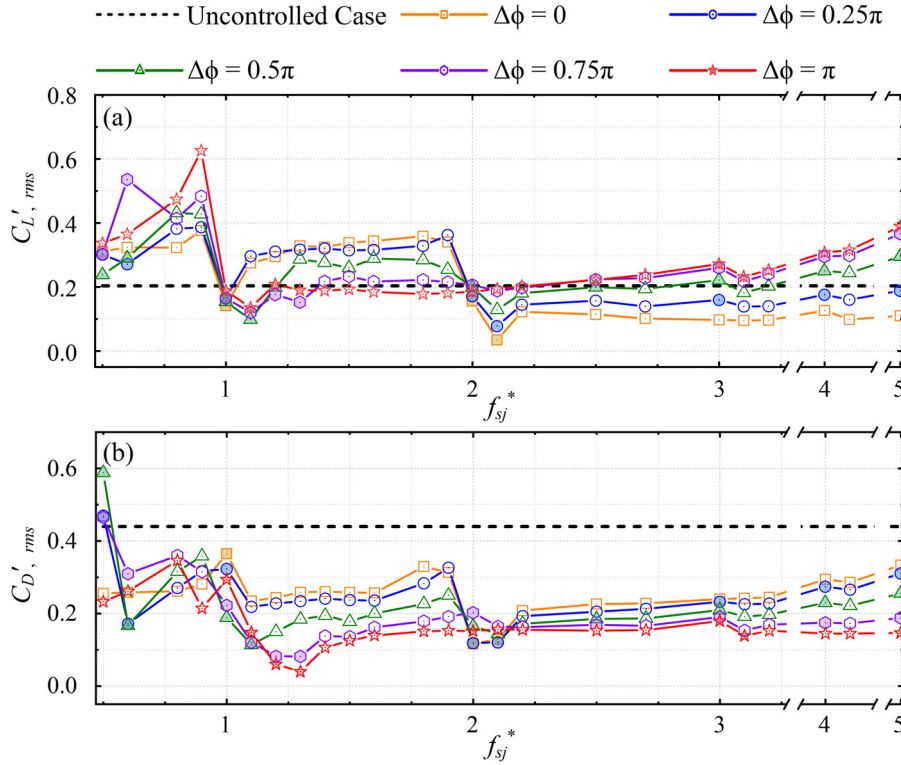


FIG. 8. Variation of (a) lift force coefficients $C'_{L,rms}$ and (b) drag coefficients $C'_{D,rms}$ with various f_{sj}^* and $\Delta\phi$. The case in the lock-on regime is denoted by the solid symbol, and the non-lock-on regimes by empty symbols. The RMS values are calculated as $C'_{L,rms} = (C_L - \bar{C}_L)_{rms}$ and $C'_{D,rms} = (C_D - \bar{C}_D)_{rms}$.

$$F_V = A_{C_v} \cos(2\pi f_{C_v} t), \quad (12)$$

where A_{C_v} and f_{C_v} are the amplitude and frequency of fluctuating vortex force, respectively. The analytical solution of Eq. (7) is obtained as

$$y(t) = \frac{A_{C_v} \cos(2\pi f_{C_v} t)}{K(1 - f_{C_v}^*)}, \quad (13)$$

where the terms related to initial conditions of undamped natural vibrations are ignored. Based on the relationship expressed in Eq. (13), the displacement is anti-phase with the vortex force when $f_{C_v} > 1$ and decreases with the augment of f_{C_v} . Thus, it can be concluded that the suppression of VIV can come up with the augment of vortex force or the higher than 1 of the vibration frequency. Although only a single-frequency harmonic is assumed here, this explanation is still valid because the vortex force can be decomposed into multiple harmonics. Considering the vortex force in Eq. (5) is part of the overall lift force, it is reasonable to believe that the variation of $C'_{L,rms}$ is not completely consistent with that of A_{C_v} . It also indicates that the suppression of VIV by SJs is not achieved by directly compelling external force in the vibrating direction.

The frequency spectrum is also calculated for more than 30 cycles based on the fast Fourier transform (FFT) and shown in Fig. 9. A single frequency is dominant in the frequency spectrum of the cylinder's motion. As expected, the cylinder vibrates at the natural frequency in the non-lock-on regime but at a frequency deviating from the natural frequency in the lock-on regime. In Fig. 9(e1), f_{sj}^* is dominant in the primary lock-on region, and $1/3$ of it dominates in the tertiary lock-on region. For the secondary lock-on region, the dominant frequency is $1/2$ of f_{sj}^* in Fig. 9(a1). Under the nonlinear effect of SJs, the frequency

spectra of hydrodynamics in lock-on or non-lock-on regimes exhibit significant differences. For the non-lock-on case, the frequency spectrum includes the natural frequency, f_n , the frequency of SJs, f_{sj} , their sum-frequency, $f_{sj} + f_n$, their difference frequency, $f_{sj} - f_n$, and harmonic frequencies. In comparison, only the frequency of SJs and its harmonic frequencies occur when lock-on happens. As shown in Fig. 9(d2), f_{sj}^* dominates the frequency spectrum when $f_{sj}^* = 1.3$, but extra components of $f^* = 0.5$ and $f^* = 1$ appear when $f_{sj}^* = 1.5$. This multi-frequency characteristic for the non-lock-on regime finally leads to different downstream wake patterns. On the contrary, the vortex shedding in the lock-on regime is more regular. Considering that fluctuating hydrodynamic is an important mechanism of the sound generation, its different frequencies can directly produce various wavelengths of sound waves. Moreover, the phase differences also induce significant differences in the frequency spectra of hydrodynamics. The proportion of high-frequency components increases as $\Delta\phi$ approaches π . However, for the part where $\Delta\phi$ is close to 0, the low-frequency component gradually dominates.

The effects of SJs on the vibration characteristics, hydrodynamics, and flow structure of the cylinder with different control parameters have been discussed. In fact, the present findings are basically consistent with the conclusions of the previous investigations.^{27,28} However, one of the advantages is that the present numerical simulation excluded the interference of reflected noise, which provides more solid evidence.

B. Effect of SJs on sound fields undergoing VIV

For the stationary circular cylinder, the vortex sheds from the upper and lower sides can generate positive and negative pulses, which

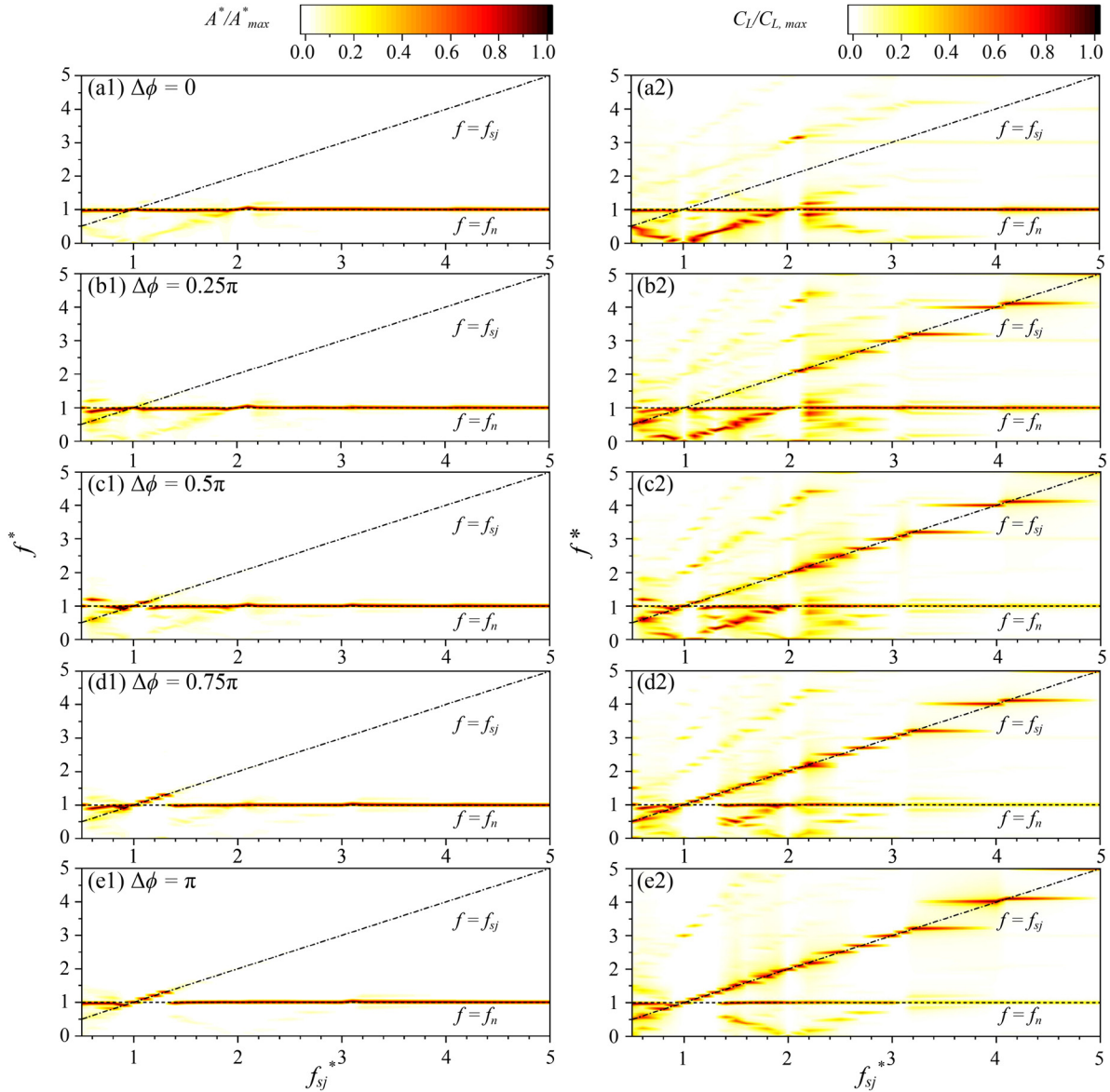


FIG. 9. Frequency spectra of (a1)–(e1) vibrating displacement A^* and (a2)–(e2) lift force coefficient C_L on various f_{sj}^* and $\Delta\phi$. For each case, both A^* and C_L are normalized by the maximum value. The dashed and dashed-dotted lines represent $f = f_n$ and $f = f_{sj}$, respectively.

propagate outward to form the sound field dominated by the lift dipole. It is also clear that the far-field sound radiation will amplify when the cylinder is vibrating.⁴ However, the effect of SJs with various f_{sj}^* and $\Delta\phi$ on the sound generation still needs to be further evaluated. To assess the component of sound sources, multi-pole decomposition of fluctuating pressure is utilized. The fluctuating pressure is defined as $p' = p - \bar{p}$, where $\bar{p}$ denotes the time-average pressure. It should be noted that the pressure is normalized by $\rho_0 c_0^2$ in the present study. Through the multi-pole decomposition, p' is described as

$$\begin{aligned}
 p'(r', \theta, t) = & A_0(r', \theta, t) \\
 & + A_1(r', \theta, t) \cos(\theta) + B_1(r', \theta, t) \sin(\theta) \\
 & + A_2(r', \theta, t) \cos(2\theta) + B_2(r', \theta, t) \sin(2\theta) \\
 & + \dots,
 \end{aligned} \tag{14}$$

where $r' = r(1 - Ma \cos \theta)$ is to consider the Doppler effect at low Mach numbers and set as $r' = 100D_0$. The terms on the right-hand side A_0 , (A_1, B_1) , and (A_2, B_2) are known as the monopole, dipole, and quadrupole, respectively. The detail of multi-pole decomposition could be found in the Ref. 6. Note that the present discussion is based on the

stage when the flow field is fully developed. The sound radiation is represented by the dilatation $\dot{p} = \partial p / \partial t$ ⁵⁴ and depicted in Fig. 10(a) with normalizing by $\frac{\rho_0 c_s^2 U_0}{D_0}$. Positive and negative pressure pulses are generated and propagate to the far field. This characteristic of sound radiation is similar to the stationary cylinder, which is also dominated by the lift dipole.

When a nonzero mass flow rate exists at a point located in the flow field, an extra monopole is generated. The presence of the monopole has great influence on far-field sound radiation. For the present case where the sound wave is much longer than the cylinder's diameter, the cylinder can be considered as a compact sound source. The strength of the monopole can be evaluated based on the overall mass flow rate on the cylinder surface from Eq. (10). Hence, the present discussion is divided into three scenarios: with the monopole, without the monopole in the non-lock-on regime, and without the monopole in the lock-on regime. Typical cases of the first two scenarios are plotted in Fig. 11, and the last scenario is discussed later. In these figures, the acoustic radiation directivity is denoted by the RMS value of p'_{rms} , which is also measured at $r' = 100D_0$ by 256 observer points in Fig. 1. Both $\dot{p}$ and p'_{rms} are calculated for more than 30 vibrating cycles.

As shown in Figs. 11(d)–11(f), the sound radiation in the lift direction exhibits the obvious asymmetry. In the lock-on regime, the monopole generated by SJs and the lift direction dipole have the same frequency. When these coherent sound waves superimpose, their phase relationship critically modulates p'_{rms} . For instance, during alternating positive/negative pulse generation from the upper/lower cylinder surfaces, the monopole radiation phase determines whether constructive or destructive interference occurs. If the monopole emits positive sound waves coinciding with positive dipole pulses, p'_{rms} becomes amplified through constructive interference, while negative dipole pulses experience cancellation through destructive interference. The interference between monopole and dipole ultimately manifests the asymmetric acoustic radiation directivity in the lift direction. In addition, when $\Delta\phi = 0.25\pi$, 0.5π , and 0.75π , the loss of spatial and temporal symmetry in the wake pattern might be another reason.

First, we consider the cases in the non-lock-on regime, as shown in Fig. 12. The discrepancy in far-field sound radiation is much larger than the near-field flow structure in Fig. 7. In Fig. 12(b) $\Delta\phi = 0.5\pi$ and 12(c) $\Delta\phi = 0$, the monopole generated from SJs dominates the

sound field. In contrast to Fig. 12(a), $\Delta\phi = \pi$, significantly intense sound is generated from the cylinder and propagates uniformly in all directions. The fluctuating pressure is decomposed into monopole, dipole, and quadrupole, which is depicted in Fig. 12. The instantaneous sound generated from anti-phase SJs is dominated by the dipole represented by the red dashed line in both lift and drag directions. While SJs with the other $\Delta\phi$ are dominated by the monopole represented by the purple dotted line. At this point, the far-field sound cannot be directly associated with variations for VIV and hydrodynamics in the presence of monopole.

Anti-phase SJs could avoid this disadvantage, as shown in Fig. 12(a1), where positive and negative pressure pulses radiate non-continuously in all directions. This also indicates that the monopole is mainly generated due to the nonzero mass flow induced by SJs. The sound wave has the different wavelength in the far field, and the instantaneous fluctuating pressure at the observer point is also not periodic. As discussed before, it is related to the multi-frequency state for non-lock-on cases. The far-field sound is generated from the cylinder's vibration and fluctuating hydrodynamics with various frequencies, as shown in Fig. 9. Sound waves propagate with different wavelengths, resulting in the irregularity of far-field sound radiation. Compared with the uncontrolled case in Fig. 10, the sound radiation in the lift direction is significantly weakened, and the strong sound waves are distributing to the drag direction. The sound reduction in the lift direction stems from the effective suppression of vortex shedding and VIV. As illustrated in Fig. 9(e2), the C_L spectrum exhibits two prominent frequency components ($f^* = 1.0$ and $f^* = 1.4$), indicating that SJs dynamically interfere with large-scale vortex formation. Specifically, Fig. 5(c) demonstrates that the natural vortex shedding is suppressed by SJs, forming vortices of different sizes, which are rapidly dissipated during downstream convection. It is also found that the frequency of fluctuating pressure in the drag direction is higher than that in the lift direction, which is different from being the same frequency when the monopole exists. Moreover, even for the non-lock-on regime with quasi-periodic fluid structures in Fig. 7, there is still no obvious drag quadrupoles represented by A'_2 from the result of multi-pole decomposition.

For the lock-on regime, we first discuss cases with the existence of the monopole. As shown in Fig. 13, it is similar to non-lock-on

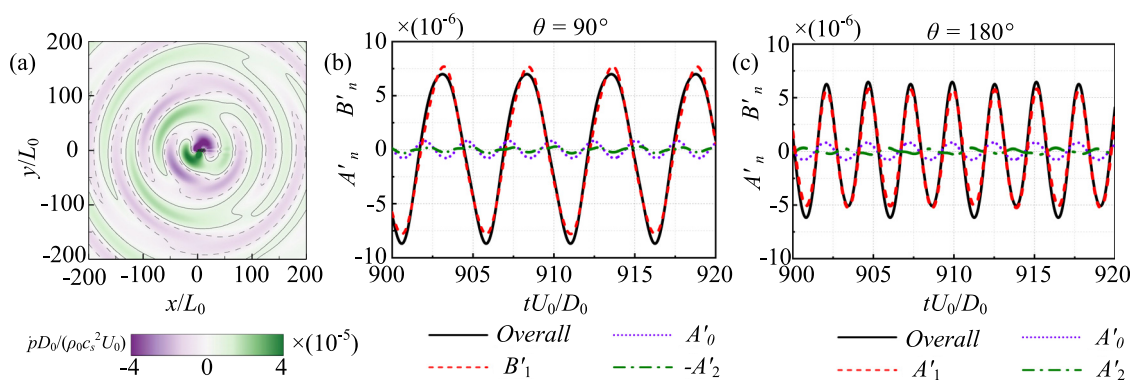


FIG. 10. (a) Instantaneous sound field for the uncontrolled case is at the time instance which $y(t)$ varies from the negative to positive and represented by dilatation and isolines with $\pm 5 \times 10^{-6}$. The second (b) and third (c) columns are results of the multi-pole decomposition on the instantaneous fluctuating pressure in the lift and drag directions, respectively.

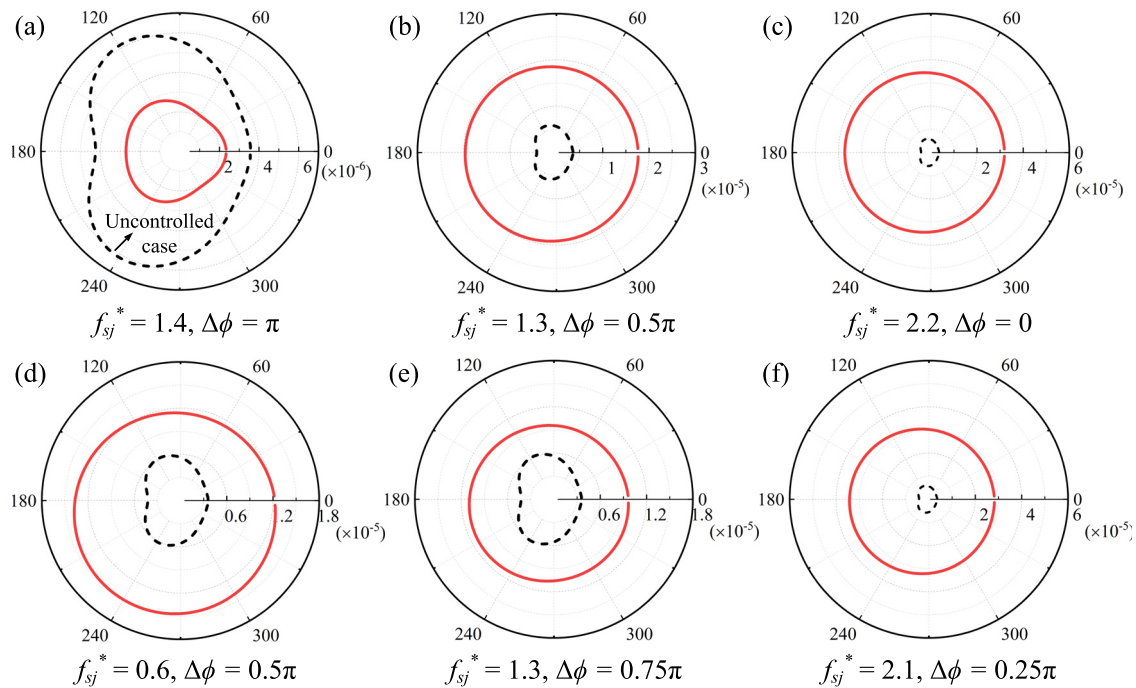


FIG. 11. The acoustic radiation directivity is represented by p'_{rms} , which is measured at $r' = 100D_0$. Cases are in (a)–(c) the non-lock-on regime and (d)–(f) the lock-on regime. The uncontrolled case is denoted by the dashed line.

cases, where the monopole dominates the far-field sound radiation. The sound waves generated by SJs propagate uniformly to all directions, and the characteristics of sound radiation are still independent of the vibrating state and the hydrodynamic variation. The wavelength of the sound wave is determined by f_{sj}^* , while the magnitude of sound is related to both f_{sj}^* and $\Delta\phi$ of SJs. At the same f_{sj}^* , SJs with $\Delta\phi$ closer to 0 produce stronger sound waves, as shown in Figs. 12(b1) and 13(b1). As discussed before, when the cylinder is regarded as a compact sound source, the sound generated by anti-phase SJs described in Eq. (10) can cancel each other out. In contrast, the sound radiation increases when $\Delta\phi$ is close to 0. The effect of f_{sj}^* can refer to Figs. 12(b1) and 13(a1). When $\Delta\phi$ is the same, SJs with high frequency can generate stronger noise. In addition, for the case with $f_{sj}^* = 1.3$ and $\Delta\phi = 0.75\pi$, the drag dipole is stronger than the lift dipole due to the suppression of VIV by SJs.

As we observed in instantaneous fluctuating pressure contours, the far-field sound radiation exhibits the monopole characteristics, and observer points at different directions receive almost the same sound wave except for the case $\Delta\phi = \pi$. More importantly, the result indicates that although SJs successfully suppressed VIV in most cases, the presence of monopole all enhances the sound radiation. In Fig. 11, the far-field sound radiation dominated by monopole gradually increases when f_{sj}^* increases or $\Delta\phi$ approaches 0. However, for the anti-phase case, the sound radiation in all directions is reduced compared to the uncontrolled case, and the sound in the drag direction is stronger than that in the lift direction. One conclusion to be drawn here is that if one would like to reduce the noise, anti-phase SJs are preferred to avoid generating extra monopoles.

Primary and tertiary lock-on regimes appear when anti-phase SJs are implemented. A significant difference also appears in the acoustic radiation directivity, as shown in Fig. 14. In both primary and tertiary lock-on regimes, the sound radiation in the lift direction reduces as A^* decreases.

Compared with the case $f_{sj}^* = 1$, A^* in the tertiary lock-on region is reduced by SJs, but the lift force increases at the same time, which may lead to intense sound radiation. However, in the drag direction, the sound radiation increases almost with the augment of f_{sj}^* . The same trend can also be observed in the time history of fluctuating pressure in Fig. 15. The acoustic radiation directivity is symmetric about the centerline. The drag dipole dominates the far-field sound radiation when A^* is adequately suppressed. The acoustic radiation directivity with high-frequency SJs is similar to the uncontrolled case. Both drag and lift dipoles appear, and the lift dipole is dominant. These results confirm the tight relationship between suppressing VIV and reducing the sound generated from dipoles.

It can also be found that the cylinder with SJs in the tertiary lock-on region radiates more intense sound in Fig. 15. The instantaneous sound field is similar to the uncontrolled case when $f_{sj}^* = 1.1$. As shown in Figs. 15(a2) and 15(a3), both the drag dipole and the lift dipole exist, where the later one has a slightly larger amplitude. Since the frequency of the lift dipole is half compared to that of the drag dipole, Fig. 15 exhibits that sound waves with larger wavelengths in the lift direction are connected to those with smaller wavelengths in both upstream and downstream. However, at $f_{sj}^* = 1.3$, the lift dipole is weakened due to the significant suppression of VIV, but the drag dipole is strengthened. The far-field sound radiation is dominated by the drag dipole. As for high-frequency SJs in Fig. 15(c1), the frequency

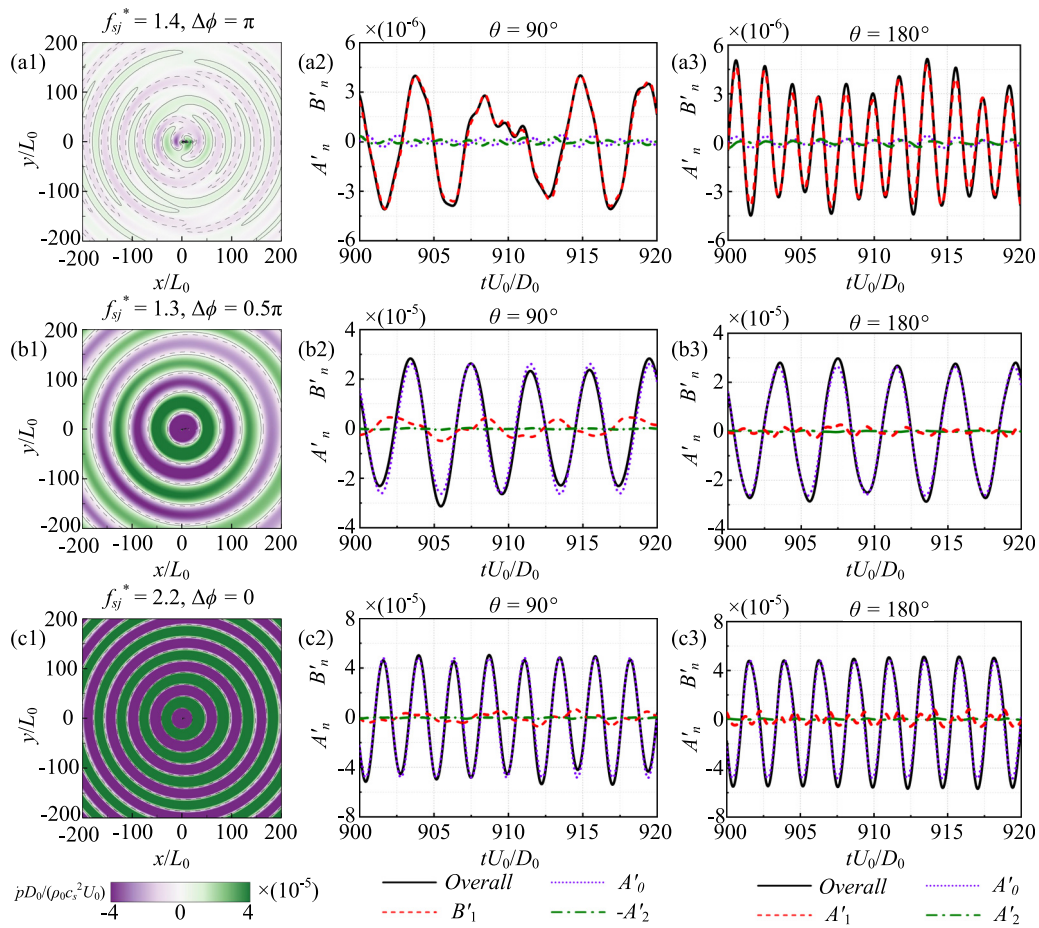


FIG. 12. (a1)–(c1) Instantaneous sound fields for non-lock-on cases are represented by dilatation. The second (a2)–(c2) and third (a3)–(c3) columns exhibit results of the multipole decomposition on the instantaneous fluctuating pressure in the lift and drag directions, respectively. By setting $\theta = 90^\circ$ and $\theta = 180^\circ$, we can obtain the time history of monopole A_0 , the lift dipole B_1 , the drag dipole A_1 , and the quadrupole A_2 according to Eq. (14).

of the lift dipole is still half that of the drag dipole, but the wavelength of sound wave is significantly shorter compared to low-frequency SJs. Moreover, due to the harmonic enhancement caused by high-frequency SJs, the time history of the sound wave is not dominated by a single frequency, which is different from the cases in the primary lock-on region.

Comparing Figs. 12(a2) and 15(c2), it can be found that the sound generated from the cylinder in the lock-on regime is stronger than that in non-lock-on and both exceeds the uncontrolled case. This phenomenon is similar to the result of Wang *et al.*,²⁸ who reported that the occurrence of lock-on may even lead to the enhancement of VIV. The direct impact of the lock-on regime is that it changes the frequency of the sound source and lead to the modulation of the sound wavelength. Here, we focus on the lock-on cases. Intensities of monopoles, drag dipoles, and lift dipoles are denoted by $A'_{0,rms}$, $A'_{1,rms}$, and $B'_{1,rms}$, respectively, and depicted in Fig. 16. As expected, $A'_{0,rms}$ is enhanced as f_{sj}^* increases or $\Delta\phi$ approaches 0. The monopole does not exist when $\Delta\phi = \pi$. For most cases, SJs weaken the dipole. At the lowest A^* when $f_{sj}^* = 1.3$ and $\Delta\phi = \pi$, the intensity of lift dipole is

reduced close by 85% compared with that of the uncontrolled case. The minimum value of $B'_{1,rms}$ occurs at $f_{sj}^* = 1.1$ and $\Delta\phi = 0.5\pi$, where the drag dipole is reduced close by 75%. All the above-mentioned results demonstrate the excellent performance of SJs in alleviating the dipole.

Since observing the displacement and the force exerting of the cylinder is much easier than measuring the far-field sound radiation, it is more practical to evaluate sound indirectly by hydrodynamics and the cylinder's vibration. Finding the connection between changes in the cylinder's vibration and hydrodynamics with far-field sound radiation is important for fully evaluating how well SJs work at reducing noise. However, it is clear that variations of hydrodynamic and the sound sources' intensity exhibit to be markedly different in Figs. 8 and 16. In the drag direction, a typical example is in the primary lock-on region, where $C_{D,rms}$ with the anti-phase SJs decreases as the augment of f_{sj}^* , but the sound radiated in the drag direction is enhanced. In the lift direction, the variation of $B'_{1,rms}$ with f_{sj}^* is much larger than the change of $C'_{L,rms}$ or A^* , especially for high-frequency SJs. The reason for this difference will be discussed in the later section.

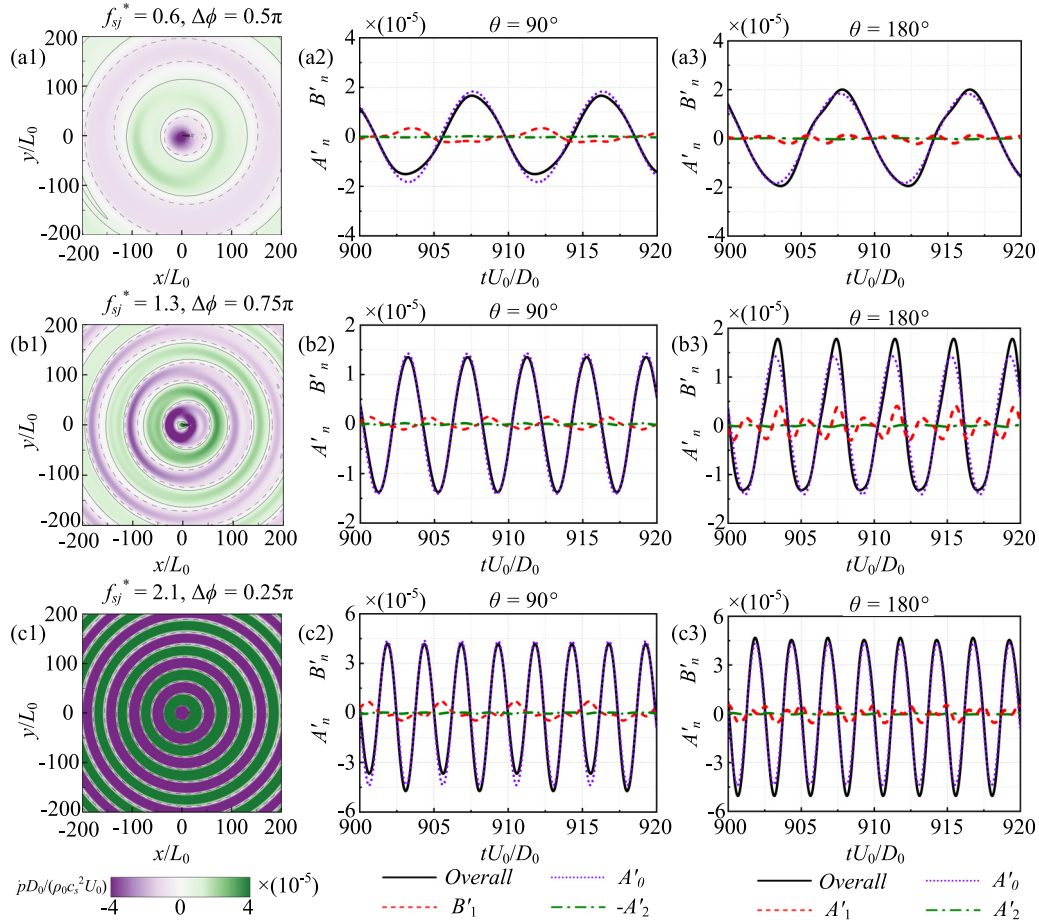


FIG. 13. (a1)–(c1) Instantaneous sound fields for lock-on cases are represented by dilatation. The second (a2)–(c2) and third (a3)–(c3) columns are results of the multi-pole decomposition on the instantaneous fluctuating pressure in the lift and drag directions, respectively.

C. Dynamic mode decomposition of fluctuating pressure fields in the lock-on regime

Dynamic mode decomposition (DMD)⁵⁵ is an effective tool for the dimensionality reduction and analysis of flow structures.⁵⁶ Since

the DMD mode contains the information for a single frequency, this advantage enables the DMD method to be applied to extract the spatial distribution of far-field sound corresponding to different frequencies. Close to $N_{DMD} = 2000$ snapshots with the data of fluctuating pressure are employed for DMD analysis. Through DMD, the fluctuating pressure can be expressed as

$$p'(\mathbf{x}, t) \approx p'_{DMD}(\mathbf{x}, t) = \sum_{i=1}^{N_{DMD}} b_i e^{i\lambda_i t} \Phi_i(\mathbf{x}), \quad (15)$$

where the instantaneous fluctuating pressure field is described by temporal characteristics $e^{i\lambda_i t}$ and spatial modes $\Phi_i(\mathbf{x})$. The growth rates and frequencies of each mode could be obtained from the real and imaginary parts of λ_i , respectively. The letter b_i is the amplitude of DMD mode. The energy of the i th mode is defined as $\sum_{j=1}^{N_{DMD}} \sum_{q=1}^{N_{Node}} (p'_{i,DMD}(q, j))^2$. It should be noted that the input data of DMD is downsampling from the instantaneous flow field by skipping one every two points for speeding up. Details of the DMD implementation in the present study could be found in Refs. 55 and 56.

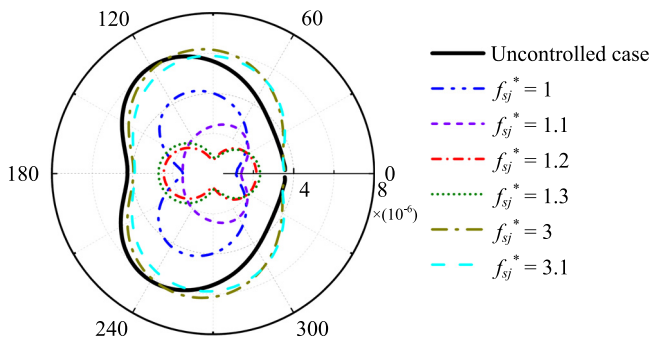


FIG. 14. For anti-phase SJs, the variation of the acoustic radiation directivity represented by p'_{rms} measured at $r' = 100D_0$ with different f_{sj}^* .

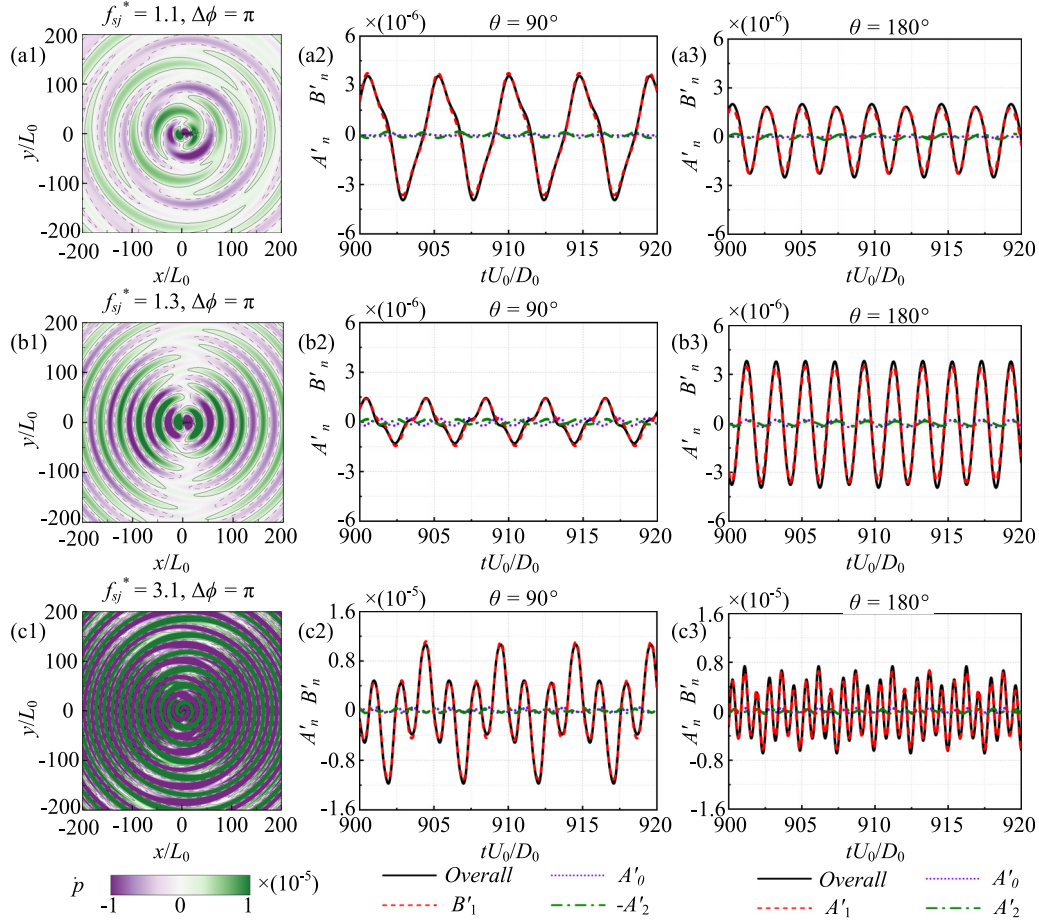


FIG. 15. For anti-phase SJs, $\Delta\phi = \pi$, (a1)–(c1) instantaneous sound fields for lock-on cases are represented by dilatation. The second (a2)–(c2) and third (a3)–(c3) columns are results of the multi-pole decomposition on the instantaneous fluctuating pressure in the lift and drag directions, respectively.

DMD analysis is conducted for the cylinder undergoing VIV, and DMD modes are depicted in Fig. 17. Energies described by p' are mainly concentrated in the first two modes, corresponding to $f^* = 1.0$ and $f^* = 2.0$, respectively. These two modes exhibit clear characteristics of lift and drag dipoles, and the wavelength of sound waves in the lift direction is twice that in the drag direction. This is consistent with the hydrodynamic variation in Fig. 3(b). For the compressible flow, the flow field contains three components: the hydrodynamic mode, the acoustic mode, and the thermal mode. In the present numerical simulation under the isothermal condition, the thermal mode could be ignored. The hydrodynamic mode convects and diffuses in the downstream region at a similar magnitude compared to the incoming flow velocity. The acoustic modes propagate at the speed of sound to the far field. The fluctuating pressure of the hydrodynamic mode adjacent to the cylinder might be more intense than that of the acoustic mode. Decomposition of hydrodynamic and acoustic modes in the flow field is another valuable topic. Such as the Doak's decomposition⁵⁷ can be utilized from the perspective of the solenoidal and irrotational components, but it is out of the scope of the present investigation. Since the DMD analysis is implemented in overall fluctuating pressure p' and

then sort DMD modes based on their energy ratios, some modes dominated by hydrodynamic modes will inevitably appear, such as $f^* = 3.0$ and $f^* = 4.0$ in Figs. 17(a4) and 17(a5). The limitation is that it is impossible to rigorously evaluate the energy of DMD modes, which completely belong to acoustic modes. It should be noted that the energy ratios of DMD modes do not strictly correlate with sound source strengths, as p' inherently weight hydrodynamic fluctuations more heavily than sound radiation.

For the case in non-lock-on regime, as shown in Fig. 17(b1), primary modes include $f^* = 1.4$, $f^* = 1.0$, $f^* = 0.4$, and $f^* = 2.8$. Moreover, in addition to the mode represented by the solid red circle, modes with relatively low energies are denoted by empty circles and distribute at various frequencies. It is consistent with the multi-frequency characteristic of the flow field in the non-lock-on regime. The cylinder vibrates at its natural frequency, which can be seen in Fig. 9. For non-lock-on cases, the frequency spectrum of the lift force shows both the natural frequency and SJs' frequency. The first two modes obtained from DMD analysis correspond to two frequencies mentioned above, $f^* = 1.4$ and $f^* = 1.0$, respectively. These two modes exhibit the characteristic of the lift dipole, while the mode $f^* = 2.8$ is similar to a

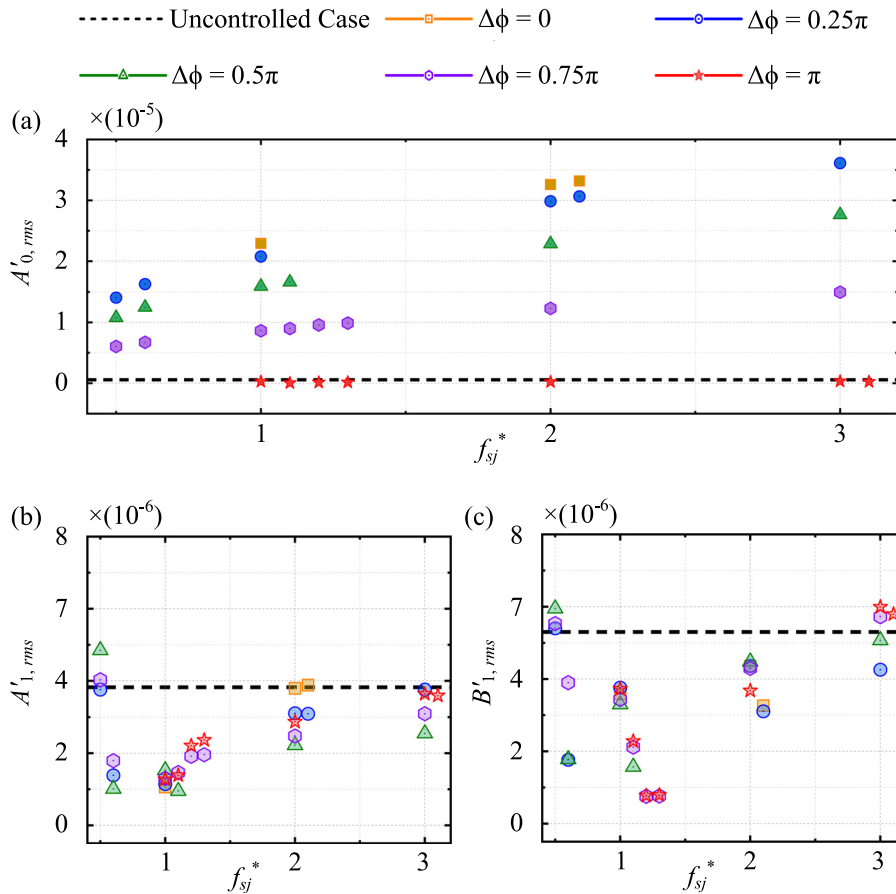


FIG. 16. In the lock-on regime, variations of the RMS value with various f_{sj}^* and $\Delta\phi$ on different sound sources: (a) the monopole A'_0 , (b) the drag dipole A'_1 , and (c) the lift dipole B'_1 . The RMS values are all calculated for more than 30 VIV cycles.

drag dipole. Model 4, corresponding to the difference frequency between SJs' frequency and the natural frequency, $f^* = 0.4$, only distributes in the downstream region adjacent to the cylinder.

Figure 17(c1) shows the result of the lock-on case, $f_{sj}^* = 2.1$ and $\Delta\phi = 0.25\pi$, in which SJs also generate monopole. It can be found that the high-energy mode is corresponding to $f_{sj}^* = 2.1$. Moreover, the fluctuating pressure in mode 1 radiates to the far field as a monopole. This result confirms that the monopole is generated from SJs with nonzero mass flow rate rather than VIV or fluctuating hydrodynamics. The vibration frequency is shifted to $f^* = 1.05$ as shown in Fig. 9. The sound radiation is similar to the uncontrolled case, which exhibits the characteristic of the lift dipole. Moreover, an indeterminate mode is also found with SJs implemented. As shown in Figs. 17(b5) $f_{sj}^* = 2.8$ and 17(c4) $f_{sj}^* = 4.2$, there always exists a dipole with the frequency of $2f_{sj}^*$ at the drag direction, whether the cylinder is or not in the lock-on regime. Considering that the frequency of the drag dipole generated by the hydrodynamic or vibration in the drag direction is always twice that of the lift direction rather than varying with SJs, which indicates that this drag dipole is likely to be generated by SJs.

For anti-phase SJs in lock-on regime, the modes with the largest energy are all corresponding to the vibration frequency because no extra monopole is introduced. In the primary lock-on region, most of the energy is concentrated in the first two modes, which is similar to the uncontrolled case. Under SJs, the vibration frequency of the cylinder

deviates from the natural frequency, and the wavelength of the sound wave for the dominant mode decreases with the augment of f_{sj}^* . However, the sound in the drag direction represented by mode 2 does not reduce with the decrease in $C'_{D,rms}$ in Fig. 8, but instead being enhanced with increasing f_{sj}^* . This difference also consists of the observations in Figs. 15(a1) and 15(b1). In Fig. 18, mode 2 of high-frequency SJs is $f^* = 3.1$ and exhibits the characteristic of the lift dipole. Since SJs are acting along the drag direction, this mode should correspond to the harmonic caused by SJs in the frequency spectrum of the lift force. However, mode 2 is generally corresponding to the frequency of drag for cases in the primary lock-on region, which indicates the different effect on the sound radiation for high-frequency SJs. In the drag direction, the dipole with $2f_{sj}^*$ also appears in addition to $f_{sj}^* = 2.06$. Moreover, in terms of the region with $100D_0 < r < 200D_0$, mode 4 is stronger than mode 3.

D. Mechanisms of sound generated from the vibrating cylinder with SJs

In this section, we will discuss the reasons for the self-noise generation from SJs and the disagreement in variations of hydrodynamic and sound radiation based on the acoustic analogy theory. The acoustic analogy approach is first proposed by Lighthill⁵⁸ and can be embedded as a post-process tool for predicting the far-field sound with low

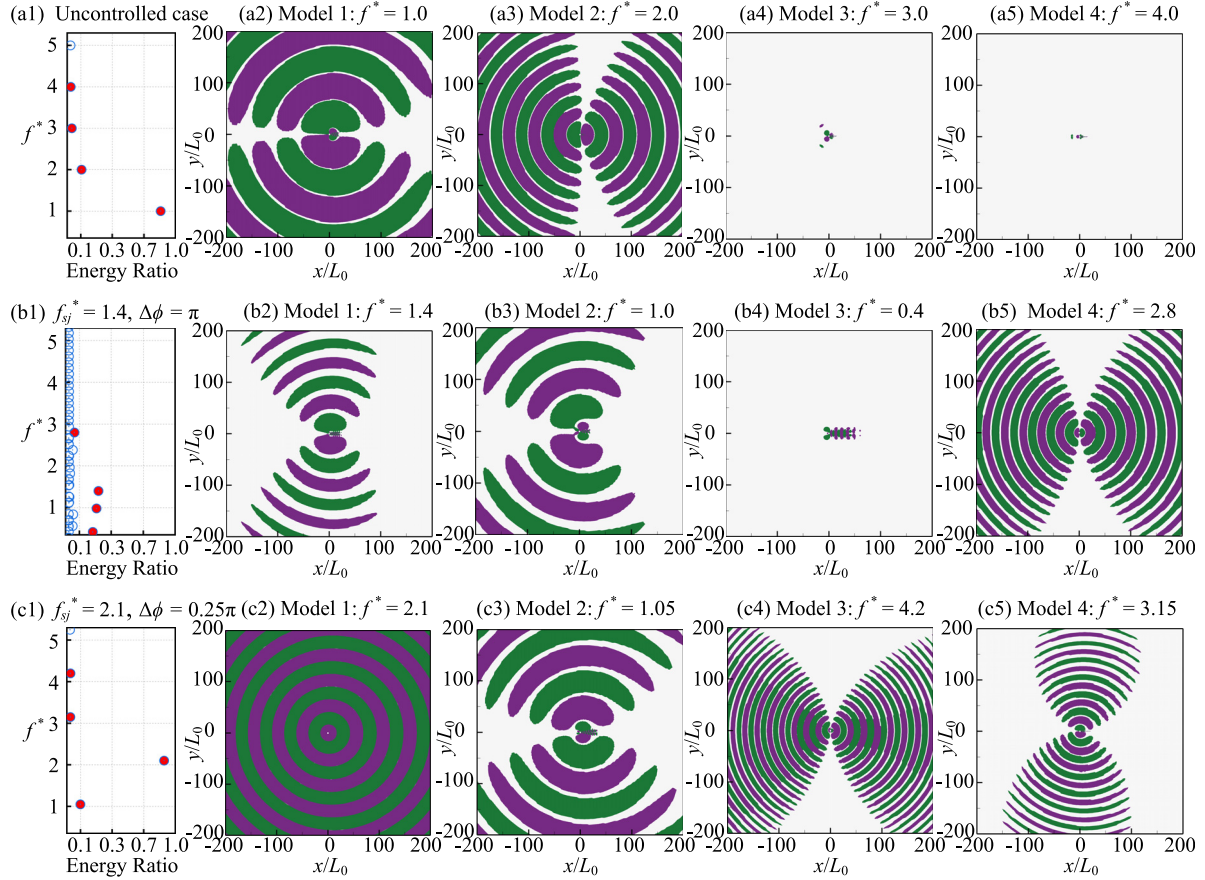


FIG. 17. Energy ratios and DMD modes for (a1)–(a5) the uncontrolled case, (b1)–(b5) $f_{sj}^* = 1.4$, $\Delta\phi = \pi$, and (c1)–(c5) $f_{sj}^* = 2.1$, $\Delta\phi = 0.25\pi$. In (a1)–(c1), the first four modes are denoted by solid red circles, and the other modes are empty circles.

computational costs. Considering the effect of moving solid boundaries, Ffowcs Williams and Hawkins⁵⁹ subsequently develop the acoustic analogy equation with the three homogeneous terms: monopole, dipole, and quadrupole. However, these three-dimensional acoustic analogy equations cannot be directly applied to two-dimensional problems. Guo⁶⁰ interpreted the different mechanisms of sound radiation in two-dimensional acoustic computation. In his work, the Ffowcs Williams and Hawkins (FW-H) equation was also developed for two-dimensional problems in the frequency domain. Similarly, Inoue and Hatakeyama⁶ developed the two-dimensional Curle's solution in the time domain by replacing the Green function. In our previous work, the monopole generated from SJs is further considered.³⁶ Following Inoue and Hatakeyama,⁶ we retained the term of monopole, and the two-dimensional Curle's solution is developed as

$$p'(\mathbf{x}, t) = \frac{c_0^{1/2}}{2^{3/2}\pi|\mathbf{r}|^{1/2}} \int_{-\infty}^0 \left[\frac{\partial}{\partial \tau} \int_S \rho_0 \mathbf{U}_n(s, \tau + \tau') dl(S) \right] \frac{d\tau'}{\sqrt{-\tau'}} + \frac{\mathbf{r}}{2^{3/2}\pi c_0^{1/2}|\mathbf{r}|^{3/2}} \int_{-\infty}^0 \left[\frac{\partial}{\partial \tau} \int_S \mathbf{f}(s, \tau + \tau') dl(S) \right] \frac{d\tau'}{\sqrt{-\tau'}}, \quad (16)$$

where S represents the cylinder surface and $\tau = t - |\mathbf{r}|/c_0$ is the delay time. The letter r is the distance between positions of observer points

and the sound source. $t' = \tau' + \tau$ represents the sound wave from the previous moments to consider the two-dimensional effect. The first and second terms on the right-hand side are the sound generated by the monopole and the dipole, respectively. $U_n = \mathbf{u} \cdot \mathbf{n}$ is the normal velocity, where $\mathbf{u}$ and $\mathbf{n}$ are the velocity vector and normal vector on the surface of the cylinder, respectively. The loading vector per unit area $\mathbf{f}$ is defined as

$$\mathbf{f} = -(\mathbf{p} + \rho_0 \mathbf{u} \mathbf{u}) \cdot \mathbf{n}, \quad (17)$$

where $\mathbf{p}$ is the tensor including the normal and shear stress. If the sound source is compact at the present low Mach number, Eq. (16) can be numerically solved as follows: First, integrate $\mathbf{f}$ and U_n along the cylinder's surface; Second, execute the time derivation in delay time τ ; Finally, integrate along the interval time τ' .

It should be noted that solving Eq. (16) cannot provide an accurate solution consistent with the present direct simulation because the cylinder is supposed to be stationary in Eq. (16) rather than moving. Nevertheless, Eq. (16) can still help us to understand the mechanism for the sound generation. When the cylinder surface is impenetrable, the monopole represented by the first term in Eq. (16) is disappeared due to $\int_S \rho_0 \mathbf{U}_n dl(S) = 0$. However, for most cases, the mass of fluid injected from upper and lower SJs cannot be completely offset, except

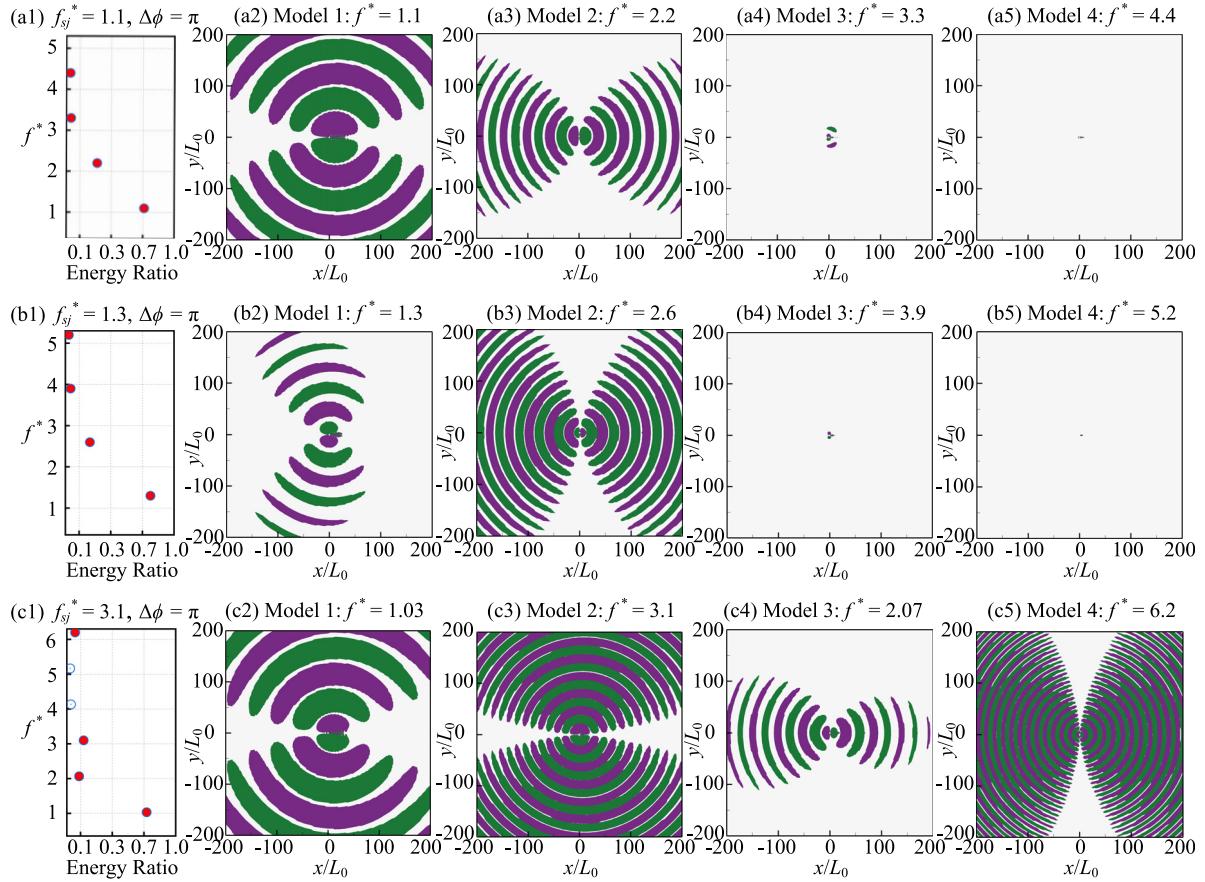


FIG. 18. Energy ratios and DMD modes with anti-phase SJs of (a1)–(a5) $f_{sj}^* = 1.1$, (b1)–(b5) $f_{sj}^* = 1.3$, and (c1)–(c5) $f_{sj}^* = 3.1$. The first four modes are also denoted by solid red circles, and the other modes are empty circles in (a1)–(c1).

when $\Delta\phi = \pi$. According to Eq. (16), this nonzero mass flow rate can finally result in the generation of the monopole. Another self-noise brought by SJs is demonstrated by Eq. (17). In addition to the fluctuating hydrodynamic, the injection of momentum flux, $(\rho_0 \mathbf{u}\mathbf{u}) \cdot \mathbf{n}$, also produces the dipole. Since SJs act on the drag direction and their velocities have been given by Eq. (10), it can be inferred that a drag dipole with the frequency of $2f_{sj}^*$ will appear. This mechanism also interprets what has been observed in the DMD analysis, namely, that the drag dipole with the frequency of $2f_{sj}^*$ always appear whether the cylinder is in the lock-on regime or not.

Moreover, results of direct acoustic simulation also exhibit that SJs with higher frequencies tend to radiate more intense noise. In fact, it is well known that the higher frequency component in hydrodynamics generates stronger sound sources, but quantitative analysis in two-dimensional conditions has not been conducted. Take the sound in the drag direction as an example, assuming the drag force in the lock-on regime is expressed as

$$F_D = A_{D1} \cos(2\pi f_{D1} t) + A_{D2} \cos(2\pi f_{D2} t + \phi_1), \quad (18)$$

where A_{D1} and A_{D2} are the amplitudes of these two harmonics. And lower and higher frequencies are represented by f_{D1} and f_{D2} , respectively. ϕ_1 is the phase difference between these two components. For

convenience, the anti-phase SJs without monopole are considered. Based on integration by substitution, the analytical solution of Eq. (16) can be obtained according to the Fresnel integral as

$$p'(\mathbf{x}, t) = \frac{r}{2c_0^{1/2} |r|^{3/2}} \left[f_{D1}^{1/2} A_{D1} \cos\left(2\pi f_{D1} \tau + \frac{\pi}{4}\right) + f_{D2}^{1/2} A_{D2} \cos\left(2\pi f_{D2} \tau + \phi_1 + \frac{\pi}{4}\right) \right]. \quad (19)$$

Since these harmonics are orthogonal, the $F_{D,rms}$ and p'_{rms} can be simplified to

$$F_{D,rms} = A_{D1,rms} + A_{D2,rms},$$

$$p'_{rms} = \frac{f_{D1}^{1/2}}{2c_0^{1/2} |r|^{1/2}} \left[A_{D1,rms} + \left(\frac{f_{D2}}{f_{D1}}\right)^{1/2} A_{D2,rms} \right]. \quad (20)$$

In Eq. (20), if the proportion of high-frequency components for $F_{D,rms}$ is $A_{D2,rms}/A_{D1,rms}$, this proportion for p'_{rms} is expanded by a factor of $(f_{D2}/f_{D1})^{1/2}$. As a result, it is clear that the high-frequency components of hydrodynamics can generate more intense noise. Again, this solution derives from the two-dimensional Curle's solution and cannot accurately describe the far-field sound radiation for present

cases with the cylinder's oscillation. Since SJs, especially with higher f_{sj}^* , often induce amounts of harmonics as shown by the frequency spectra in Fig. 9, it is also confirmed as an important reason for the deterioration of noise reduction performance.

IV. SUMMARY AND CONCLUSION

The control of aeolian tones from the circular cylinder undergoing VIV by a pair of synthetic jets (SJs) is numerically investigated in the laminar condition. SJs with various frequencies (f_{sj}^*) and phase differences ($\Delta\phi$) act along the inflow direction. The effects of f_{sj}^* and $\Delta\phi$ on VIV, the hydrodynamic, and the far-field sound radiation are comprehensively evaluated. Moreover, the spatial distribution and intensity of the sound corresponding to different frequencies are determined by DMD analysis. Finally, the mechanism of sound generation is analyzed based on the acoustic analogy theory and summarized in Fig. 19. The following are the key findings:

1. For most cases, SJs have successfully suppressed VIV and significantly affect the hydrodynamic. Some cases might be in the lock-on regime, and the vibration frequency deviates from the natural frequency. The present acoustic simulation could effectively eliminate the interference of reflected sound waves, which provides new evidence for the occurrence of lock-on regimes.
2. When SJs are not set as anti-phase, the monopole in the far field is dominant, and the sound radiation increases significantly compared with the uncontrolled case. The smaller intensity of the dipole makes it submerged in the monopole, which ultimately leads to the absence of evaluating the far-field sound based on the variation of the flow field.
3. The obvious effect of SJs on the intensity, the acoustic radiation directivity, and the sound wavelength in the far field is demonstrated. The occurrence of lock-on directly affects the sound wavelength but does not necessarily lead to the sound reduction. Due to the elimination of the monopole, anti-phase SJs in the primary lock-on region successfully reduce the lift dipole by a maximum of 85% and the drag dipole by 75%. This also leads the drag dipole to dominate.
4. DMD analysis of the fluctuating pressure gives the spatial distribution about the component of sound at different frequencies. The occurrence of lock-on regimes, the monopole, and the use of high-frequency SJs all have important impacts on the primary DMD modes. Furthermore, we also find that no matter in lock-on regime or not, the drag dipole with $2f_{sj}^*$ exists in each case.

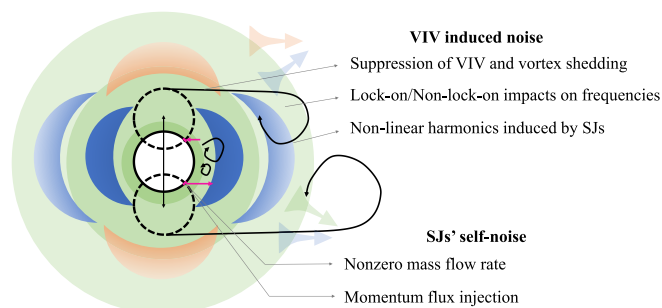


FIG. 19. Schematic diagram of the mechanism on the far-field sound radiation affect by SJs when the cylinder is undergoing VIV.

5. Based on Curle's theory, it can be concluded that the non-zero mass flow rate and momentum flux variation from SJs can generate the monopole and the drag dipole with $2f_{sj}^*$, respectively. And the further theoretical analysis demonstrates that the proportion of the high-frequency component induced by SJs is amplified by the square root of the frequency in far-field sound under the two-dimensional condition.

This research advances our understanding of how SJs affect the sound radiation from the circular cylinder undergoing VIV. The performance of SJs on sound modification is mainly determined by the suppression of VIV and the self-noise of SJs. Despite being obtained in the laminar condition, the present result can provide information on the feasibility and difficulty of noise reduction using SJs. In addition to the noise reduction, the future work will integrate energy consumption into a multi-objective optimization framework, combining flow-induced noise mechanism in the optimization algorithm to identify that maximize noise suppression while minimizing energy input. Moreover, the experimental validation in turbulent flow regimes will further refine these insights.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Jian Song: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal). **Haibao Hu:** Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal). **Feng Ren:** Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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